

Step 4: Zero-Shot Validation & Inference

Sliding-Window Volumetric Reconstruction

The False Positive Problem

Generative models hallucinate. To prevent the pipeline from mistakenly connecting disjoint neurons or background artifacts, NeuroDiffusion implements a **Three-Layered Gauntlet**.

1. **Topological Gate:** The Backward model outputs $p \approx 0$ for structurally incompatible endpoints.
 2. **Structural Gate:** The Heuristic Evaluator penalizes biologically absurd generated curves.
 3. **Physical Gate:** The Intensity Cost ensures the generated curve overlaps with bright biological signals in the raw image.
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The Validity Heuristic Evaluator (Cost A)

Trained as a binary classifier/discriminator: - **Architecture:** A 2-layer EGNN operating over the generated path sequence, concatenated with the fragment contexts. Outputs a scalar predicting the Structural Neural Gap. - **Training:** Optimized via BCE against true biological paths (Label 1) and corrupted/hallucinated paths (Label 0). - **Effect:** Heavy penalty for mathematically sound but biologically impossible structures (e.g., hairpin turns).

Biological Intensity Path Integral (Cost B)

The ultimate physical ground-truth validation during live inference over TIFF stacks.

$$\text{Cost}_B(X) = -\frac{1}{|X|} \sum_i |x_i \cdot \text{Tangent}(x_i)| \cdot \text{StepDir}(\text{Vesselness}(x_i))$$

- Evaluates if the path cuts through dark space (Low Vesselness).
 - Evaluates if the path travels perpendicular to the actual physical neurites (Tangent Misalignment).
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End-to-End Inference Architecture

Processing a 4GB TIFF volume requires VRAM-conscious engineering (inference/main.py):

- **CPU Producer:** A multi-threaded process loads overlapping sliding windows (e.g., $256 \times 256 \times 64$) into RAM and pre-computes Sato/Frangi Vesselness and Tangent volumes.
- **Multi-GPU Consumer:** Concurrently evaluates candidate endpoints inside the window using the Sequence Generator and Evaluator.
- **Topological Export:** Reconstructed connections are mapped back to micron-scale SWC graphs, crucially preserving exact parent-child topological links (rendering true continuous skeletons, not point clouds).